Proof of the Algebraic Independence of π with e and γ

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Abstract

This paper establishes a criterion for the algebraic independence of two numbers. Applying this theorem, we prove that π and the Euler–Mascheroni constant γ are algebraically independent. Consequently, it follows that $\pi\gamma$, π/γ, $\pi+\gamma$, and $\pi-\gamma$ are all transcendental. Furthermore, we prove that π and e are algebraically independent. Finally, it is proved that $\zeta(k)$, π, and γ are algebraically independent. In this paper, π denotes the Wallis product π_w, and γ denotes the Maclaurin series γ_M.

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1 Concept

[1, p. 6] The set of natural numbers is divided into two parts: one part consists of those that can be expressed by numerical expressions, referred to as expressible natural numbers; the other part consists of those that cannot be expressed by numerical expressions, referred to as inexpressible natural numbers, as shown

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below:

$$\underbrace{0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad \cdots}_{\text{expressible natural numbers}} \underbrace{N-3 \quad N-2 \quad N-1 \quad N}_{\text{inexpressible natural numbers}}$$

The upper bound of the natural numbers is N. Any number exceeding N, such as $2N$ or N^2, is no longer a natural number.

All expressible natural numbers are finite. The sum or product of two finite natural numbers remains finite. Moreover, the sum or product of finitely many finite natural numbers is also finite; that is, if a and b are finite, then $a^b < N$.

2 Theorem 1. A Sufficient Condition for Algebraic Independence

Let P be an arbitrary prime and let $f(N)$, $h(N)$ be functions of N. If two numbers S and T can be expressed as the following irreducible fractions:

$$S = P^{\pm f(N)} \cdot \frac{B}{A}, \quad \gcd(P, B) = 1,$$

$$T = P^{\pm h(N)} \cdot \frac{E}{D}, \quad \gcd(P, E) = 1,$$

where $A \neq 0$, $B \neq 0$, $E \neq 0$, and $D \neq 0$, then $f(N) \to \infty$, $h(N) \to \infty$, and $f(N)/h(N) \to \infty$ is a sufficient condition for the algebraic independence of S and T.

3 Proof of Theorem 1

The general expression for the polynomial in S and T is given by:

$$F(S, T) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_{ij} S^i T^j.$$

For any monomial $F(S, T) = aS^iT^j$ with $a \neq 0$, we have $F(S, T) \neq 0$. Henceforth, we only consider polynomials with more than one term; that is, polynomials consisting of at least two terms.

First, consider the case where $S = \dfrac{P^{f(N)}}{D}$, i.e., when $P^{f(N)}$ appears in the numerator (the $\pm$ signs of each term are chosen independently; the choice of sign at any position is independent of those at other positions):

$$F(S,T) = \sum_{i=0}^{n}\sum_{j=0}^{m} \pm a_{ij} S^i T^j.$$

Express the coefficients a_{ij} as irreducible fractions:

$$a_{ij} = \frac{P^{c_{ij}} g_{ij}}{b_{ij}} \quad \text{or} \quad \frac{g_{ij}}{P^{c_{ij}} b_{ij}},$$

where all $b_{ij}, g_{ij}, c_{ij}, n, m$ are finite natural numbers, $b_{ij} \neq 0$, $\gcd(P, g_{ij}) = 1$, and $\gcd(P, b_{ij}) = 1$.

Substituting S and T into $S^i T^j$ yields:

$$S^i T^j = \frac{P^{if(N)} B^i}{A^i} \cdot \frac{P^{\pm jh(N)} E^j}{D^j} = \frac{P^{if(N) \pm jh(N)} B^i E^j}{A^i D^j}.$$

When $P^{c_{00}}$ appears in the numerator, i.e., $a_{00} = \dfrac{P^{c_{00}} g_{00}}{b_{00}}$, adopting the conventions that $0 \cdot f(N) = 0$ and $x^0 = 1$, we have:

$$a_{ij} S^i T^j = \frac{P^{if(N) \pm jh(N) \pm c_{ij}} B^i E^j g_{ij}}{A^i D^j b_{ij}};$$

$$F(S,T) = \sum_{i=1}^{n}\sum_{j=1}^{m} \pm \frac{P^{if(N) \pm jh(N) \pm c_{ij}} B^i E^j g_{ij}}{A^i D^j b_{ij}} \pm \frac{P^{c_{00}} g_{00}}{b_{00}}.$$

After combining the terms over a common denominator and simplifying the numerator of each term $a_{ij} S^i T^j$ to the form $P^{if(N) \pm jh(N) \pm c_{ij}} G_{ij}$, where $\gcd(P, G_{ij}) = 1$, we obtain:

$$F(S,T) = \frac{\displaystyle\sum_{i=1}^{n}\sum_{j=1}^{m} \pm P^{if(N) \pm jh(N) \pm c_{ij}} G_{ij} \pm P^{c_{00}} G_{00}}{A^n D^m \prod b_{ij}}.$$

Whether $F(S,T)$ vanishes depends entirely on the numerator; the denominator need not be considered at all.

Since $c_{ij}/h(N) \to 0$ and $h(N)/f(N) \to 0$, it follows that when $i_1 > i_2$, we necessarily have:

$$i_1 f(N) \pm j_s h(N) \pm c_{uv} > i_2 f(N) \pm j_k h(N) \pm c_{wx}.$$

When $i_1 = i_2$ and $j_1 > j_2$, we necessarily have:

$$if(N) \pm j_1 h(N) \pm c_{uv} > if(N) \pm j_2 h(N) \pm c_{wx}.$$

In other words, among all the terms of the polynomial $F(S,T)$, no two terms share the same exponent of the factor P in their numerators.

Arbitrarily partition these terms into two groups, α and β, each containing at least one term. Let L_α denote the minimum exponent of P among all terms in group α, and L_β denote the minimum exponent of P among all terms in group β. Then it necessarily follows that:

$$L_\alpha \neq L_\beta.$$

Let $\mathrm{SUM}(\alpha)$ denote the sum of the numerators of all terms in group α. Then we have:

$$\mathrm{SUM}(\alpha) = \sum_{k\in\alpha} \pm P^{L_\alpha} H_{\alpha k} = P^{L_\alpha} \sum_{k\in\alpha} \pm H_{\alpha k}, \quad \gcd\left(P, \sum_{k\in\alpha} \pm H_{\alpha k}\right) = 1.$$

Let $\mathrm{SUM}(\beta)$ denote the sum of the numerators of all terms in group β. Then we have:

$$\mathrm{SUM}(\beta) = \sum_{k\in\beta} \pm P^{L_\beta} H_{\beta k} = P^{L_\beta} \sum_{k\in\beta} \pm H_{\beta k}, \quad \gcd\left(P, \sum_{k\in\beta} \pm H_{\beta k}\right) = 1.$$

By the Fundamental Theorem of Arithmetic, if two numbers have equal absolute values, their factorizations must be identical up to sign. Thus:

$$\begin{aligned} L_\alpha \neq L_\beta &\implies |\,\mathrm{SUM}(\alpha)| \neq |\,\mathrm{SUM}(\beta)| \\ &\implies \mathrm{SUM}(\alpha) \pm \mathrm{SUM}(\beta) \neq 0 \\ &\implies F(S,T) \neq 0. \end{aligned}$$

This implies that algebraic independence holds for S and T.

When $P^{c_{00}}$ appears in the denominator, i.e., $a_{00} = \dfrac{g_{00}}{P^{c_{00}} b_{00}}$, we have:

$$F(S,T) = \frac{\displaystyle\sum_{i=1}^{n}\sum_{j=1}^{m} \pm P^{if(N)\pm jh(N)\pm c_{ij}} G_{ij}}{A^n D^m \prod b_{ij}} \pm \frac{G_{00}}{P^{c_{00}} A^n D^m \prod b_{ij}},$$

$$P^{c_{00}}F(S,T) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} \pm P^{if(N)\pm jh(N)\pm c_{ij}+c_{00}}G_{ij} \pm G_{00}}{A^n D^m \prod b_{ij}}.$$

By the same method as above, we can prove the following:

$$P^{c_{00}}F(S,T) \neq 0 \implies F(S,T) \neq 0.$$

This implies that algebraic independence holds for S and T.

When $S = \dfrac{E}{P^{f(N)}D}$, i.e., when $P^{f(N)}$ appears in the denominator, the argument is simpler since the term a_{00} need not be considered. We have:

$$F(S,T) = \sum_{i=1}^{n}\sum_{j=1}^{m} \pm\frac{B^i E^j g_{ij}}{P^{if(N)\mp jh(N)\mp c_{ij}}A^i D^j b_{ij}} \pm \frac{P^{\pm c_{00}}g_{00}}{b_{00}},$$

$$F(S,T) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} \pm\frac{G_{ij}}{P^{if(N)\mp jh(N)\mp c_{ij}}} \pm P^{\pm c_{00}}G_{00}}{A^n D^m \prod b_{ij}}.$$

After combining the terms over the common denominator $P^{nf(N)\mp mh(N)\mp c_{nm}}A^n D^m \prod b_{ij}$ and simplifying the numerator of each term $a_{ij}S^i T^j$ to the form $P^{(n-i)f(N)\mp(m-j)h(N)\mp c_{nm}\pm c_{ij}}G_{ij}$, where $\gcd(P, G_{ij}) = 1$, we obtain:

$$F(S,T) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} \pm P^{(n-i)f(N)\mp(m-j)h(N)\mp c_{nm}\pm c_{ij}}G_{ij} \pm P^{nf(N)\mp mh(N)\mp c_{nm}\pm c_{00}}G_{00}}{P^{nf(N)\mp mh(N)\mp c_{nm}}A^n D^m \prod b_{ij}}.$$

By the same reasoning, it follows that $F(S,T) \neq 0$. Hence, algebraic independence holds for S and T.

4 Algebraic Independence of π and γ

Following the convention of Qi Jin, we adopt the Wallis product of π as its algebraic definition, and the Maclaurin series of γ as its definition[2, pp. 13–14]. According to the Wallis product, we have:

$$\begin{aligned}\pi &= 2\prod_{n=1}^{\infty}\frac{4n^2}{4n^2-1} = 2\prod_{n=1}^{M}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right)\\ &= \frac{2^{2M+1}\,M!\,M!}{(2M-1)!!\,(2M+1)!!}.\end{aligned}$$

Let $v_p(n)$ denote the p-adic valuation of n (i.e., the exact exponent of the prime p in the prime factorization of n). Throughout this paper, we adopt the convention that $v_2(\pi)$, $v_2(\gamma)$, and $v_2(e)$ denote the exponent of 2 in the numerator or denominator of the irreducible fraction representing π, γ, and e, respectively. By Legendre's formula, it is well known that

$$v_p(n!) = \frac{n}{p-1} + O(\log n),$$

and in particular $v_p(n!) \sim n/(p-1)$ as $n \to \infty$. Thus, we have:

$$\begin{aligned} v_2(\pi) &= v_2\big(2^{2M+1}\, M!\, M!\big) \\ &= 2M + 1 + M + M + 2\, O(\log M) \\ &= 4M + 1 + 2\, O(\log M). \end{aligned}$$

When $N = 2M + 1$, we obtain

$$v_2(\pi) = 2N - 1 + 2\, O(\log M) \to \infty.$$

When $N = 2M + 2$, we obtain

$$v_2(\pi) = 2N - 3 + 2\, O(\log M) \to \infty.$$

Based on Qi Jin's calculations ([2, pp. 6–9]; see Appendix A for details), we have:

$$v_2(\gamma) \geq (N-1)h > (N-1)\log_2\left(\frac{N}{2}\right) \to \infty.$$

Consequently, it follows that:

$$\frac{v_2(\gamma)}{v_2(\pi)} > \frac{(N-1)\log_2(N/2)}{2N - 1 + 2\, O(\log M)} \to \infty.$$

This satisfies the sufficient condition of Theorem 1; therefore, π and γ are algebraically independent. It further follows that $\pi\gamma$, π/γ, $\pi + \gamma$, and $\pi - \gamma$ are all transcendental numbers.

5 π and e Are Algebraically Independent

When $N = 2M + 1$, based on the definition of e, we have:

$$e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = \left(1 + \frac{1}{2M}\right)^{2M} = \left(\frac{2M+1}{2M}\right)^{2M}.$$

When $N = 2M + 2$, based on the definition of e, we have:

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \left(1+\frac{1}{2M+1}\right)^{2M+1} = \left(\frac{2M+2}{2M+1}\right)^{2M+1}.$$

In all cases, the numerator or denominator of e contains the factor $(2M + 1)$, with its exponent $y = N - 1$.

According to the Wallis product, we have:

$$\pi = 2\prod_{n=1}^{\infty}\frac{4n^2}{4n^2-1} = 2\prod_{n=1}^{M}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right) = \frac{2\,(2M)!!\,(2M)!!}{(2M-1)!!\,(2M+1)!!}.$$

It can be seen that the denominator of π also contains the factor $(2M + 1)$, with its exponent equal to 1.

Performing prime factorization on the odd number $(2M+1)$, since $2M+1 \to \infty$, there exist only the following three cases:

Case 1. The exponent of some factor P ($P \neq 2$) tends to infinity, i.e., $2M+1 = P^zX$, where $z \to \infty$ and $z \leq \log_P(2M + 1)$. Then we have:

$$2M - 1 = P^zX - 2, \quad \gcd(P, X) = 1.$$

Every odd number $P^k \cdot (2r - 1)$ has a corresponding even number $P^k \cdot (2r)$ paired with it, namely:

$$P^k \cdot 1 \leftrightarrow P^k \cdot 2, \quad P^k \cdot 3 \leftrightarrow P^k \cdot 4, \quad \ldots,$$
$$P^k \cdot \big(XP^{z-k} - 2\big) \leftrightarrow P^k \cdot \big(XP^{z-k} - 1\big), \quad \text{where } k \leq z.$$

It follows that the exponent of P in $(2M - 1)!!$ is equal to the exponent of P in $(2M)!!$ when considering only the case $k \leq z$, i.e.:

$$v_P\big((2M-1)!!\big) = v_P\big((2M)!!\big), \quad k \leq z.$$

Among all natural numbers, there must exist a number $P^{z_{\max}}$ such that $P^{z_{\max}} \leq 2M + 1$ and $P^{z_{\max}+1} > 2M + 1$.

When $X > P$, we have $z_{\max} > z$, and there also exist cases where $z < k \leq z_{\max}$.

If $\lfloor X/P^s \rfloor$ is even for $s = 1, 2, \ldots, z_{\max} - z$, considering only the case $k = z + s$, then the exponent of P in $(2M - 1)!!$ equals the exponent of P in $(2M)!!$.

If $\lfloor X/P^s \rfloor$ is odd, then the exponent of P in $(2M - 1)!!$ exceeds that in $(2M)!!$ by 1. Consequently, the range of the exponent of P in the irreducible fraction representation of π can be determined as follows:

$$z \leq v_P(\pi) \leq z + 2(z_{\max} - z) = 2z_{\max} - z \leq 2\log_P(2M + 1).$$

That is:

$$v_P(\pi) \geq z \to \infty.$$

Meanwhile, $v_P(e) = z(N-1) \to \infty$, and we have:

$$\frac{v_P(e)}{v_P(\pi)} \geq \frac{z(N-1)}{2\log_P(2M+1)} \geq \frac{z(N-1)}{2\log_P(N-1)} \to \infty.$$

Therefore, π and e satisfy the sufficient condition stated in Theorem 1, and hence π and e are algebraically independent.

Case 2. $(2M+1)$ has infinitely many prime factors. Since the polynomial coefficients are rational numbers, they can involve only finitely many prime factors; hence there must exist some prime factor P of $(2M+1)$ that does not appear in either the numerator or the denominator of any coefficient. Let $v_P(2M+1) = u$, where u is finite. Using the same calculation method as in Case 1, we can similarly obtain:

$$u \leq v_P(\pi) \leq u + 2(z_{\max} - u) = 2z_{\max} - u \leq 2\log_P(2M+1).$$

When $v_P(\pi) \to \infty$, as in Case 1, the sufficient condition for Theorem 1 is satisfied.

When $v_P(\pi)$ takes a finite value r, i.e., $v_P(\pi) = r$, we have:

$$v_P(\pi) = r, \quad v_P(e) = v_P\big((2M+1)^{N-1}\big) = u(N-1).$$

When a prime factor P does not occur in any of the coefficients, the sufficient condition in the theorem can be weakened to $f(N)/h(N) \to \infty$, without requiring $h(N) \to \infty$. Indeed, in the polynomial

$$F(\pi, e) = \sum_{i=0}^{n} \sum_{j=0}^{m} \pm a_{ij}\, \pi^i e^j,$$

no coefficient a_{ij} contains the factor P in its numerator or denominator. Thus (P may appear in either the numerator or the denominator of e; it can only appear in the denominator of π):

$$a_{ij}\, \pi^i e^j = X_{ij}\, P^{\pm ju(N-1) - ir}, \quad \gcd(P, X_{ij}) = 1.$$

Since u and r are finite and $N \to \infty$, it follows that:

When $j_1 \neq j_2$, we necessarily have:

$$\pm j_1 u(N-1) - i_s r \neq \pm j_2 u(N-1) - i_t r.$$

When $j_1 = j_2$ and $i_1 \neq i_2$, we necessarily have:

$$\pm ju(N-1) - i_1 r \neq \pm ju(N-1) - i_2 r.$$

In other words, the powers of P across all terms of the polynomial are distinct. By the same argument as in the proof of Theorem 1, it follows that $F(\pi, e) \neq 0$ for any non-zero polynomial F, which implies that π and e are algebraically independent.
Case 3. The factors include inexpressible infinite numbers. When $(2M+1)$ is prime, or when its prime factors include an inexpressible number—for example, $2M+1 = 3 \cdot (2M+1)/3$, where $(2M+1)/3$ is prime—let P denote this inexpressible prime factor. Then the factor P does not appear in either the numerator or the denominator of any coefficient a_{ij} of the polynomial. As in Case 2, it follows that π and e are algebraically independent.

6 Algebraic Independence of $\zeta(k)$, π, and γ

Theorem 2. The proof of Theorem 1 can be generalized to an arbitrary number of variables. Let P be an arbitrary prime and let $f_i(N)$ be functions of N. Suppose that k numbers $S_1, S_2, \ldots, S_k$ can be expressed as the following irreducible fractions:

$$S_i = P^{\pm f_i(N)} \cdot \frac{B_i}{A_i}, \quad \gcd(P, B_i) = 1,$$

where $A_i \neq 0$ and $B_i \neq 0$. Then the conditions

$$f_i(N) \to \infty \quad \text{and} \quad \frac{f_{i+1}(N)}{f_i(N)} \to \infty$$

are sufficient for the algebraic independence of $S_1, S_2, \ldots, S_k$.
Proof omitted.

By definition, $\zeta(k)$ denotes the value of the Riemann zeta function at positive integers, where k is an integer greater than 1. Qi Jin has proved that $\zeta(k)$ are all transcendental numbers. Based on Qi Jin's calculations ([2, pp. 12–14]; see Appendix B for details), we have:

$$k \log_2 N \geq v_2(\zeta(k)) > k \log_2(N/2) \to \infty.$$

Since

$$\frac{v_2(\pi)}{v_2(\zeta(k))} > \frac{2N - 3 + 2\mathrm{O}(\log M)}{k \log_2 N} \to \infty$$

and

$$\frac{v_2(\gamma)}{v_2(\pi)} \to \infty,$$

it follows from Theorem 2 that $\zeta(k)$, π, and γ are algebraically independent.

Final Remark

In this paper, π denotes the Wallis product π_w, and γ denotes the Maclaurin series γ_M. The author does not establish whether π coincides with π_w, or whether γ coincides with γ_M. The results proved in this paper are the following:

- π_w and γ_M are algebraically independent.
- π_w and e are algebraically independent.
- $\zeta(k)$, π_w, and γ_M are algebraically independent.

References

[1] Jin, Q. (2026). *Proof of the Continuum Hypothesis*, preprint. Zenodo: `https://doi.org/10.5281/zenodo.20651018`.

[2] Jin, Q. (2026). *Proof of Transcendence of the Euler–Mascheroni Constant* γ, preprint. Zenodo: `https://doi.org/10.5281/zenodo.23282062`.

Appendix

A Qi Jin's Calculation of $v_2(\gamma)$

From the definition of the Euler–Mascheroni constant, we have[2, pp. 6–9]:

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \ln n\right) = \sum_{k=1}^{N}\frac{1}{k} - \ln N$$

$$= \sum_{k=1}^{N}\frac{1}{k} - \ln\left(\frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{N-1}{N-2}\cdot\frac{N}{N-1}\right)$$

$$= \sum_{k=1}^{N} \frac{1}{k} - \sum_{k=1}^{N-1} \ln \frac{k+1}{k} = \sum_{k=1}^{N} \frac{1}{k} - \sum_{k=1}^{N-1} \ln\left(1 + \frac{1}{k}\right).$$

The Taylor expansion of $\ln(1+x)$ about $x = 0$ is given by:

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{x^N}{N}, \qquad -1 < x \leq 1.$$

Substituting the respective values yields (note that N may be either odd or even, and consecutive terms alternate in sign):

$$\ln \frac{2}{1} = \ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \pm \frac{1}{N-1} \mp \frac{1}{N};$$

$$\ln \frac{3}{2} = \ln\left(1 + \frac{1}{2}\right) = \frac{1}{2} - \frac{1}{2^2 \cdot 2} + \frac{1}{2^3 \cdot 3} - \frac{1}{2^4 \cdot 4} + \cdots \pm \frac{1}{2^{N-1}(N-1)} \mp \frac{1}{2^N N};$$

$$\ln \frac{4}{3} = \ln\left(1 + \frac{1}{3}\right) = \frac{1}{3} - \frac{1}{3^2 \cdot 2} + \frac{1}{3^3 \cdot 3} - \frac{1}{3^4 \cdot 4} + \cdots \pm \frac{1}{3^{N-1}(N-1)} \mp \frac{1}{3^N N};$$

$$\ln \frac{5}{4} = \ln\left(1 + \frac{1}{4}\right) = \frac{1}{4} - \frac{1}{4^2 \cdot 2} + \frac{1}{4^3 \cdot 3} - \frac{1}{4^4 \cdot 4} + \cdots \pm \frac{1}{4^{N-1}(N-1)} \mp \frac{1}{4^N N};$$

$$\vdots$$

$$\begin{aligned} \ln \frac{N}{N-1} &= \ln\left(1 + \frac{1}{N-1}\right) \\ &= \frac{1}{N-1} - \frac{1}{(N-1)^2 \cdot 2} + \frac{1}{(N-1)^3 \cdot 3} - \frac{1}{(N-1)^4 \cdot 4} + \cdots \\ &\quad \pm \frac{1}{(N-1)^{N-1}(N-1)} \mp \frac{1}{(N-1)^N N}. \end{aligned}$$

Among the natural numbers up to N, there necessarily exists a power of two, 2^h, satisfying

$$\frac{N}{2} < 2^h \leq N.$$

This 2^h is the unique integer in $\{1, 2, \ldots, N\}$ with the highest power of 2 in its factorization; for every other natural number in this range, the exponent of 2 in its prime factorization is at most $h - 1$. We have:

$$\begin{aligned} \ln\left(1 + \frac{1}{2^h}\right) &= \frac{1}{2^h} - \frac{1}{2^{2h} \cdot 2} + \frac{1}{2^{3h} \cdot 3} - \frac{1}{2^{4h} \cdot 4} + \cdots \\ &\quad \pm \frac{1}{2^{(N-3)h}(N-3)} \mp \frac{1}{2^{(N-2)h}(N-2)} \pm \frac{1}{2^{(N-1)h}(N-1)} \mp \frac{1}{2^{Nh} N}. \end{aligned}$$

Let $v_p(n)$ denote the p-adic valuation of n (i.e., the exact exponent of the prime p in the prime factorization of n). When $N-1 \neq 2^h$, among the denominators of all terms, the one with the highest power of 2 is $2^{Nh}N$. We have:

$$v_2(2^{Nh}N) = Nh + v_2(N).$$

Let $v_2(\text{other})$ denote the highest power of 2 in the denominators of the remaining terms. Then we have:

$$v_2(\text{other}) < Nh + v_2(N).$$

When $N-1 = 2^h$, since $\gcd(N, N-1) = 1$ and $v_2(N) = 0$, we have:

$$v_2(2^{(N-1)h}(N-1)) = v_2(2^{Nh}N) = Nh.$$

Combining the last two terms of this expansion yields:

$$\frac{1}{2^{(N-1)h}(N-1)} - \frac{1}{2^{Nh}N} = \frac{1}{2^{(N-1)h}2^h} - \frac{1}{2^{Nh}(2^h+1)}$$

$$= \frac{1}{2^{Nh}} - \frac{1}{2^{Nh}(2^h+1)} = \frac{2^h}{2^{Nh}(2^h+1)} = \frac{1}{2^{(N-1)h}(2^h+1)}.$$

After combining the last two terms of the expansion of $\ln(1+1/2^h)$, the term whose denominator contains the highest power of 2 among all terms is precisely this combined term, with denominator $2^{(N-1)h}(2^h+1)$. We have:

$$v_2(2^{(N-1)h}(2^h+1)) = (N-1)h,$$

$$v_2(\text{other}) < (N-1)h.$$

Rewrite each term in the form $\dfrac{1}{2^j X_{il}}$, where $\gcd(2, X_{il}) = 1$. There exists a unique term for which j attains its maximum value; denote this maximal exponent by $j_{\max}$ and the corresponding X_{il} by $X_{\max}$. For all remaining terms, denote the exponent by j_{other} and the corresponding odd factor by X_{other}, where $j_{\text{other}} \leq j_{\max} - 1$.

Taking $(N!)^N$ as the common denominator to combine the sum, and writing $(N!)^N = 2^T W$, with $\gcd(2, W) = 1$, we obtain:

$$\gamma = \sum \frac{1}{2^j X_{il}} = \frac{\sum \dfrac{2^T W}{2^j X_{il}}}{2^T W} = \frac{\sum \dfrac{2^{T-j} W}{X_{il}}}{2^T W}$$

$$= \frac{\dfrac{2^{T-j_{\max}}W}{X_{\max}} + \sum \dfrac{2^{T-j_{\text{other}}}W}{X_{\text{other}}}}{2^T W}$$

$$= \frac{2^{T-j_{\max}}\left(\dfrac{W}{X_{\max}} + \sum \dfrac{2^{j_{\max}-j_{\text{other}}}W}{X_{\text{other}}}\right)}{2^T W}$$

$$= \frac{\dfrac{W}{X_{\max}} + \sum \dfrac{2^{j_{\max}-j_{\text{other}}}W}{X_{\text{other}}}}{2^{j_{\max}} W}$$

$$= \frac{\dfrac{W}{X_{\max}} + 2\sum \dfrac{2^{j_{\max}-j_{\text{other}}-1}W}{X_{\text{other}}}}{2^{j_{\max}} W}.$$

Since $j_{\max} - j_{\text{other}} - 1 \geq 0$, we have:

$$\gcd\left(2,\, 2\sum \frac{2^{j_{\max}-j_{\text{other}}-1}W}{X_{\text{other}}}\right) = 0.$$

As $\gcd(2, W/X_{\max}) = 1$, it follows that:

$$\gcd\left(2,\, \frac{W}{X_{\max}} + 2\sum \frac{2^{j_{\max}-j_{\text{other}}-1}W}{X_{\text{other}}}\right) = 1.$$

Consequently, the factor $2^{j_{\max}}$ in the denominator of γ cannot be canceled.

Since $j_{\max} = Nh + v_2(N)$ or $(N-1)h$, we have $f(N) \geq (N-1)h \to \infty$, which satisfies the sufficient condition for transcendental numbers. Therefore, γ is both transcendental and irrational.

B Qi Jin's Calculation of $v_2(\zeta(k))$

The primitive definition of $\zeta(k)$ is the value of the Riemann zeta function at positive integers. The method used to prove the transcendence of γ can similarly be applied to prove that all $\zeta(k)$ are transcendental numbers[2, pp. 12–14].

$$\zeta(k) = \sum_{n=1}^{\infty} \frac{1}{n^k}, \qquad k > 1.$$

Specifically,

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{N^3},$$
$$\zeta(7) = \sum_{n=1}^{\infty} \frac{1}{n^7} = \frac{1}{1^7} + \frac{1}{2^7} + \frac{1}{3^7} + \cdots + \frac{1}{N^7},$$
$$\zeta(m) = \sum_{n=1}^{\infty} \frac{1}{n^m} = \frac{1}{1^m} + \frac{1}{2^m} + \frac{1}{3^m} + \cdots + \frac{1}{N^m}.$$

By using $(N!)^m$ as a common denominator, and writing $(N!)^m = 2^T W$, with $\gcd(2, W) = 1$, we obtain:

$$\begin{aligned}\zeta(m) &= \frac{\frac{(N!)^m}{1^m} + \frac{(N!)^m}{2^m} + \cdots + \frac{(N!)^m}{(N-1)^m} + \frac{(N!)^m}{N^m}}{(N!)^m} \\ &= \frac{\frac{2^T W}{1^m} + \frac{2^T W}{2^m} + \cdots + \frac{2^T W}{(N-1)^m} + \frac{2^T W}{N^m}}{2^T W}.\end{aligned}$$

Among the natural numbers up to N, there necessarily exists a power of two, 2^h, satisfying

$$\frac{N}{2} < 2^h \leq N.$$

This 2^h is the unique integer in $\{1, 2, \ldots, N\}$ with the highest power of 2 in its factorization; for every other natural number in this range, the exponent of 2 in its prime factorization is at most $h - 1$. Let the largest such exponent be denoted by $h_{\max}$, and the powers of 2 in the other terms be denoted by h_{other}. Then we have:

$$h_{\max} - h_{\text{other}} \geq 1.$$

Hence $\zeta(m)$ can be expressed as:

$$\begin{aligned}\zeta(m) &= \frac{2^{T - m h_{\max}} W + \sum \frac{2^{T - m h_{\text{other}}} W}{X_{\text{other}}}}{2^T W} \\ &= \frac{2^{T - m h_{\max}} \left(W + 2^{m(h_{\max} - h_{\text{other}})} \sum \frac{W}{X_{\text{other}}} \right)}{2^T W} \\ &= \frac{W + 2^{h_{\max} - h_{\text{other}}} \sum \frac{W}{X_{\text{other}}}}{2^{m h_{\max}} W}.\end{aligned}$$

The numerator is odd, and the factor $2^{mh_{\max}}$ in the denominator cannot be cancelled out. Since $N/2 < 2^{h_{\max}} \leq N$, it follows that:

$$h_{\max} > \log_2\left(\frac{N}{2}\right) \quad \Longrightarrow \quad mh_{\max} > m\log_2\left(\frac{N}{2}\right) \to \infty.$$

Therefore, all $\zeta(k)$ are transcendental numbers, and consequently, irrational numbers.